Yu Wang

AI Engineer / Data Scientist | GenAI, RAG & Evaluation

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Profile

AI engineer and recent Computer Science Ph.D. with hands-on experience building Python systems for retrieval-augmented generation, structured LLM extraction, and evaluation-driven ML research. Strong in designing experiments, retrieval pipelines, and human-in-the-loop workflows, with peer-reviewed publications and cross-functional collaboration across research and product settings.

Experience

Sep 2025–Present **Software Engineer**, *i-Refact*, Den Bosch, Netherlands

- Built FastAPI, SQLAlchemy, PostgreSQL, and Pydantic services for a versioned schema-editing and data-modelling platform used in regulated financial-reporting workflows.
- Implemented workspace, publishing, and conflict-resolution flows; maintained OpenAPI contracts, migrations, and Pytest coverage for critical backend logic.
- Designed an AI workflow that converts expert conversations and internal documentation into structured domain models, and implemented the transcript-to-conceptual-model stage as a Python prototype using schema-constrained LLM outputs, evidence grounding, and graph rendering.

Mar 2023–Jul 2025 **Ph.D. Researcher**, *Utrecht University*, Utrecht, Netherlands

- Authored multiple peer-reviewed publications at the intersection of interpretable ML, data visualization, and applied AI; won the Best Student Paper Award at IVAPP 2024.
- Designed experiments, evaluation protocols, and reusable Python and JavaScript research code for explaining classifier behaviour and analysing high-dimensional data.
- Developed a controllable inverse-projection method in PyTorch with adversarial training and interactive tools for human-in-the-loop exploration of high-dimensional image and latent-space data, in collaboration with supervisors and external researchers.

Feb 2022–Sep 2022 **AI Engineer Intern**, *Schlumberger Technologies Ltd.*, Beijing, China

- Built Python workflows over heterogeneous drilling and geomechanics data for digital geostructure modelling and automated formation labelling.
- Reduced a 3D formation-labelling algorithm from $O(n^3)$ to $O(n^2)$, cutting runtime from hours to minutes through domain-informed constraints.
- Developed GAN-based lithology reconstruction experiments and interactive 3D visualizations; earned 2nd place in the PUTC Data Science Hackathon 2022.

Selected GenAI Projects

RAG Thesis Assistant — Built a FastAPI application around my Utrecht University Ph.D. thesis using chunking, ChromaDB-backed hybrid retrieval, optional cross-encoder reranking, citation lookup, and short-term conversation memory for grounded question answering.

Transcript Knowledge Extraction — Built a Python prototype using Instructor, Gemini, and Pydantic to extract entities, relationships, assumptions, and open questions from transcripts, with evidence grounding and optional graph visualization for human review.

LCIP — Designed and implemented a controllable inverse-projection system in PyTorch for exploring high-dimensional image data and StyleGAN latent spaces from 2D embeddings, combining generative modelling with interactive visual analytics.

Education

2023–2025 **Ph.D. in Computer Science**, *Utrecht University*, Netherlands
Thesis: Enhanced decision maps for exploring classification models

2019–2025 **Ph.D. in Geology**, *China University of Geosciences, Beijing*, China
Applied machine learning and visualization to mineral genesis classification

Skills

Python / Backend	Python, FastAPI, SQLAlchemy, PostgreSQL, Alembic, Pydantic, pytest, OpenAPI, Docker, Kubernetes
LLM / Retrieval	RAG, chunking, embeddings, cross-encoder reranking, prompt design, structured outputs, vector search, retrieval evaluation
ML / Data	pandas, NumPy, scikit-learn, XGBoost, PyTorch, TensorFlow, GANs, interpretable ML, experiment design
Libraries	ChromaDB, LangChain, Instructor, PydanticAI
Visualization	Matplotlib, seaborn, D3.js, PySide, pyqtgraph, VisPy
Languages	Mandarin (native), English (working proficiency), Dutch (A1)

Recognition

Awards	Best Student Paper Award, IVAPP 2024; 2nd Prize, PUTC Data Science Hackathon 2022

Selected Publications

2026	Wang, Y., Dennig, F., Behrisch, M., Telea, A. *LCIP: Loss-Controlled Inverse Projection of High-Dimensional Image Data*. arXiv:2602.11141.
2025	Wang, Y., Grosu, C., Telea, A. *Computing fast and accurate maps for explaining classification models*. Computers & Graphics, 109, 104230.
2024	Wang, Y., Qiu, K., Telea, A., Hou, Z., Zhou, T., Cai, Y., Ding, Z., Yu, H., Deng, J. *Interpreting mineral deposit genesis classification with decision maps: A case study using pyrite trace elements*. American Mineralogist.
2024	Wang, Y., Telea, A. *Fundamental Limitations of Inverse Projections and Decision Maps*. Proc. IVAPP, 571–582.
Open source	Accepted visualization fix contribution to the SHAP project (#2302).
More	See Google Scholar for the full publication list.

Selected Talks

2024	Speaker, International Conference on Information Visualization Theory and Applications (IVAPP), Rome
2023	Speaker, EGU General Assembly, Vienna
2020	Speaker, National Conference on Mineral Deposits, Hangzhou